

Anomalous Aortic Origin of a Coronary Artery Repair Through an Anterior Minithoracotomy

Innovations

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Abstract

Objective: The benefits of minimally invasive adult cardiac surgery are well established. Nevertheless, minimally invasive congenital cardiac procedures, even for adult patients, are uncommon. In 2018, we started repairing anomalous aortic origin of a coronary artery (AAOCA) through a 5 cm anterior minithoracotomy when possible to improve cosmesis and avoid sternal precautions. We hypothesized this approach was safe and reliable. **Methods:** A 5 cm incision was made in the right second intercostal space. The incision was carried down to the pericardium while preserving the internal mammary artery. With the pericardium in view, the second and third ribs were disarticulated. Central cardiopulmonary bypass was established, and the repair was carried out based on the patient's anatomy. The technique was modified to a left anterior minithoracotomy for 1 patient who required pulmonary artery translocation. At any point, if the dissection or repair was not progressing appropriately, the minimally invasive exposure was converted to a partial or traditional median sternotomy. **Results:** Between June 2018 and June 2019, 11 patients underwent minimally invasive anomalous coronary repair. Four patients (3 with body mass index >30) were converted to traditional sternotomy due to poor visualization. Postoperatively, 1 patient required coronary artery bypass after 335 days, due to extensive collaterals and stable angina. Otherwise, at a median follow-up of 437 days (IQR 340 to 480), patients had resumed baseline activity without recurrent symptoms. **Conclusions:** Minimally invasive AAOCA repair may be appealing, although surgeons should be cautious given the high conversion rate.

Central Message

Anomalous aortic origin of a coronary artery repair through a 5 cm anterior minithoracotomy is associated with a high conversion rate, particularly for patients with body mass index >30 and those requiring coronary reimplantation.

Keywords

congenital heart disease, minimally invasive surgery, anomalous coronary, anomalous aortic origin of coronary artery

Introduction

Advancements in minimally invasive approaches have transformed adult cardiac surgery over the last 20 years.^{1–3} Despite the reduced surgical trauma, postoperative pain, and potentially shorter lengths of stay, the field of adult congenital cardiac surgery has been slow to adopt these techniques.^{2,3}

Anomalous aortic origin of a coronary artery (AAOCA) is a rare, although potentially lethal condition. The majority of these anomalies are asymptomatic, yet they represent the second most common cause of sudden cardiac death in young athletes.⁴ Traditionally, AAOCA is repaired through a median sternotomy.^{5–7}

Starting in 2018, we introduced a minimally invasive 5 cm right anterior minithoracotomy to repair AAOCA.⁸ The

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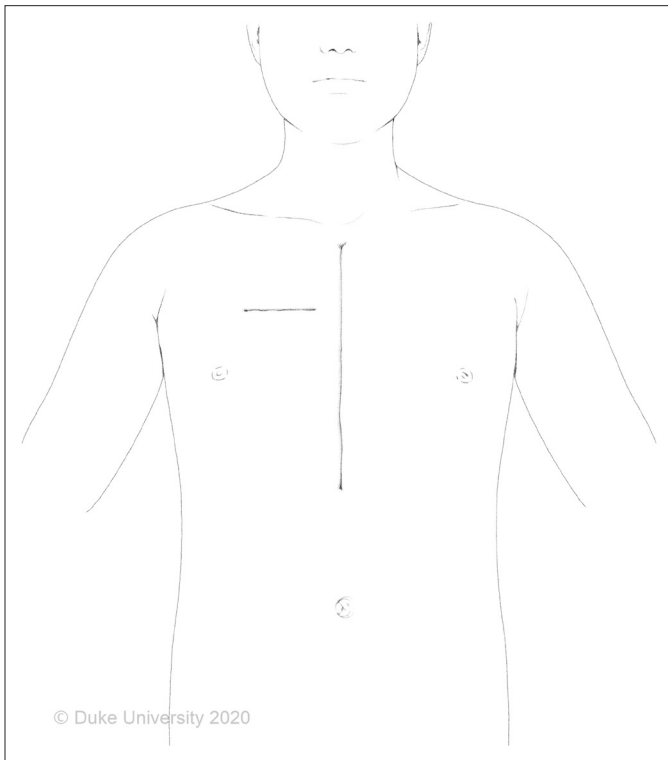


Fig. 1. Minimally invasive anomalous aortic origin of a coronary artery approach; 5 cm right anterior minithoracotomy versus traditional median sternotomy at scale.

motivation for this approach was to improve cosmesis and avoid sternal precautions. The following case series outlines our experience in 11 patients during our first year including anomalous left (LCA) and right coronary artery (RCA) unroofing, neo-ostial creation, coronary reimplantation, and pulmonary artery translocation.

Surgical Technique

Patients

Institutional Review Board approval was given to review patients who underwent AAOCA repair through a minithoracotomy between June 2018 and June 2019. Eleven patients were identified, and a chart review was conducted to gather information related to their preoperative, operative, and postoperative course.

Preoperative variables included patient age, sex, body mass index (BMI), symptoms, and coronary anatomy. Operative variables included date of operation, type of repair performed, whether the minithoracotomy approach was aborted, and total cross-clamp, cardiopulmonary bypass, and operative times. Postoperative variables included nights in the intensive care unit (ICU), days in the hospital (LOS), procedural complications, pain scores (on postoperative day [POD] 1, 3, and at discharge), morphine equivalents (ME; on POD 1, 3, and

discharge), days to latest follow-up, interval imaging results, and symptoms.

Pain was calculated by averaging the patient reported pain scores over a 24 hr period (0700 to 0700) based on a 10-point scale (zero being no pain and 10 being the most pain). The denominator for the calculation was the number of reported pain scores within the 24 hr period. Opioids received on POD 1, 3, and discharge were converted to ME received over a 24 hr period (0700 to 0700).⁹

Surgical Assessment

Prior to discussing operative repair of AAOCA, patient clinical history and imaging were reviewed. Patients presenting with AAOCA involving the left coronary were always offered surgery, whereas patients with right AAOCA needed to be symptomatic without an alternative etiology for their symptoms before surgery was offered. Work-up started with echocardiography, and AAOCA anatomy was confirmed with computed tomography angiography (CTA). Given the low risk of sudden cardiac death for patients with right AAOCA, exercise stress test was performed if the history or imaging studies were equivocal.¹⁰ As previously described, AAOCA with an intramural course is approached through a transverse aortotomy.^{11,12} Under direct visualization, the decision was made to unroof the intramural segment if it was above the commissure or, in the case that it traversed below the commissure, create a neo-ostium in the distal intramural course in the correct sinus (neo-ostial creation). If the AAOCA ostium was diminished, no intramural course was identified, and there was sufficient surrounding aortic tissue, coronary reimplantation to the physiologic coronary sinus was performed. The AAOCA involving a common ostia and no intramural course was treated with a lateral pulmonary artery translocation.

Surgical Approach

The chest was prepped and draped in standard fashion. A median sternotomy was marked on the patient, and the sternal saw prepared in case emergent conversion was needed. A 5 cm incision was made adjacent to the sternum in the right second intercostal space (Fig. 1). The incision was carried down to the pericardium in a lateral to medial fashion, being careful to preserve the internal mammary artery. With the pericardium in view, the second and third ribs were disarticulated, and a small rib retractor was used to maintain the field. The pericardium was opened sharply along the greater curve of the aorta and extended inferiorly. Silk sutures (3-0) were used to create a pericardial cradle, exposing the superior vena cava, right atrium, and ascending aorta (Fig. 2). A soft tissue wound protector was placed to improve the exposure provided by the pericardial cradle. Selective lung ventilation was not required to provide necessary exposure.

After the patient was heparinized, central cardiopulmonary bypass was established. The aorta was cannulated near the

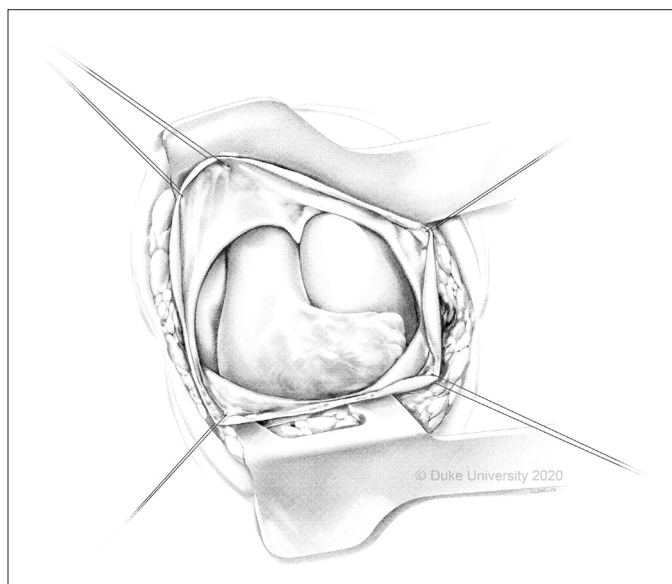


Fig. 2. Minimally invasive anomalous aortic origin of a coronary artery exposure. With the pericardium opened, the superior vena cava and right atrium are seen on the left. The ascending aorta is on the right of the figure. The internal mammary artery is preserved during this approach.

proximal arch or superior aspect of the exposure, so that the cross clamp could be placed approximately 3 to 4 cm above the sinotubular junction. The bypass cannulas were retracted

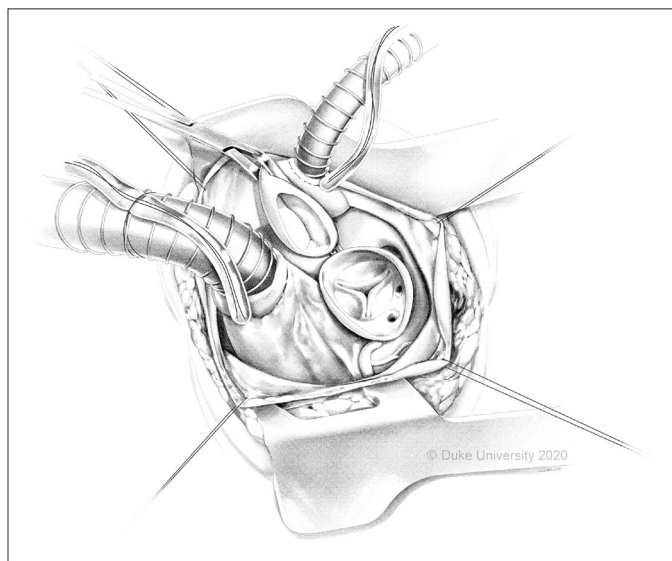


Fig. 3. Cannulation strategy and aortotomy demonstrating anomalous aortic origin of a coronary artery anatomy. Cardiopulmonary bypass is established centrally and the aortotomy is made 1 cm superior to the sinotubular junction. In this case, an anomalous left coronary artery is seen originating from the right coronary sinus. The left ventricle is also vented intraoperatively through transaortic drop suction which is not depicted here in order to provide a clearer visual.

radially away from the operative field. Once on bypass, antegrade cardioplegia was given to arrest the heart.

The aorta was opened through a transverse aortotomy 1 cm above the sinotubular junction (Fig. 3). Although not shown, the left heart was vented directly through the aortic valve. The course of the AAOCA was probed with coronary dilators, and the indicated repair was then carried out based on the patient's anatomy. If the dissection or repair was not satisfactory, the minimally invasive approach was converted to a partial or traditional median sternotomy. Conversion was possible within minutes without alterations to the established cannulation strategy. Following repair, the aorta was closed in 2 layers and the patient was weaned off bypass. The repair was assessed with transesophageal echocardiography. If satisfactory, the bypass cannulas were removed, a 24 F mediastinal Blake drain was placed, and the minithoracotomy was closed in layers. The patient was then extubated in the operating room and transported to the ICU where they received nitroglycerin for 24 hr and aspirin thereafter.

The technique was modified to a left anterior minithoracotomy for 1 patient who required pulmonary artery translocation. This patient was cannulated for bypass peripherally through his femoral artery and vein, the heart was emptied, and the main pulmonary artery was relocated laterally along the left pulmonary to prevent compression of the AAOCA.

Results

Eleven patients underwent AAOCA repair through an anterior minithoracotomy between June 2018 and June 2019 (Table 1). Patients were all symptomatic on presentation. The most common symptoms were angina (9 patients, 82%), syncopal or presyncopal episodes (8 patients, 73%), and palpitations (3 patients, 27%), whereas one patient presented after an ST-segment elevation myocardial infarction. CTA was used to diagnose AAOCA in all patients, and 6 patients underwent preoperative exercise stress tests to confirm the etiology of the symptoms. None of the patients had a prior median sternotomy.

Seven patients (64%) successfully had their AAOCA repaired through an anterior minithoracotomy. Four patients (3 with BMI >30) were converted to a partial or complete median sternotomy due to inadequate exposure. The median LOS in the intensive care unit was 1 night regardless of conversion (IQR 1 to 2 nights; Table 2). For patients who were not converted, the total LOS was 5 days (IQR 4 to 5.5 days), and the ME at the time of discharge were zero (IQR 0 to 8.5). Those undergoing sternotomy spent 6.5 days in the hospital (IQR 6 to 8.25 days) and required 15 ME at the time of discharge (IQR 11.25 to 19.25).

At a median follow-up time of 437 days (IQR 340 to 480), there were 2 notable complications. One patient who previously underwent neo-ostial creation required subsequent coronary artery bypass for stable angina nearly a year after discharge. Previous intraoperative and postoperative

Table 1. Preoperative and Intraoperative Characteristics of Patients Undergoing Anterior Minithoracotomies for AAOCA.

Patient	Age	Sex	BMI	Year of procedure	Procedure performed	Converted	Cross-clamp time (min)	CPB time (min)	Operative time (min)
1	52	M	44	2018	RCA unroofing	PS	55	86	317
2	27	F	24	2018	RCA unroofing	—	57	76	180
3	37	M	33	2018	RCA reimplantation, CABG	PS	142	266	505
4	15	M	20	2018	PA translocation	—	-	112	286
5	52	F	27	2018	RCA unroofing	—	63	75	176
6	14	M	23	2018	RCA unroofing	—	45	68	256
7	30	M	25	2019	RCA neo-ostial creation	—	130	180	314
8	47	F	26	2019	RCA unroofing, reimplantation, CABG	MS	172	296	545
9	15	M	21	2019	LCA unroofing	—	92	136	285
10	19	M	47	2019	RCA unroofing	—	74	133	321
11	57	F	39	2019	LCA unroofing	MS	133	159	344
Median (IQR), all patients							83 (59-132)	133 (81-170)	314 (270-332)
Median (IQR), excluding those converted							69 (59-88)	112 (77-134)	285 (201-300)

Abbreviations: AAOCA, anomalous aortic origin of a coronary artery; BMI, body mass index; CABG, coronary artery bypass grafting; CPB, cardiopulmonary bypass; F, female; LCA, left coronary artery; M, male; MS, median sternotomy; PS, partial sternotomy; RCA, right coronary artery.

echocardiography had demonstrated significant collaterals and a patent neo-ostia with satisfactory flow. Although follow-up cardiac catheterization demonstrated progressive collateralization of the proximal RCA, competitive flow, and occlusions in segments 1 and 2, necessitating ligation and bypass on POD 335. Another patient who was converted to median sternotomy for inadequate exposure developed a sternal wound infection that ultimately required operative debridement on POD 23. Otherwise, there was no evidence of phrenic nerve injury or musculoskeletal impairment in the remaining patients at the time of their last follow-up. Six patients (54%) had CTA or

exercise stress echocardiography performed during their documented follow-up. Aside from the patient who underwent CABG on POD 335, these studies were all negative for stenosis or inducible ischemia.

Discussion

While AAOCA repair through an anterior minithoracotomy is possible, surgeons should be cautious due to a high conversion rate. Our minimally invasive AAOCA approach was originally intended to improve cosmetic outcomes and avoid

Table 2. Postoperative Characteristics of Patients Undergoing Anterior Minithoracotomies for AAOCA.

Patient	ICU nights	Hospital nights	Complications	POD 1 pain	POD 1 total ME	POD 3 pain	POD 3 total ME	DC pain	DC total ME
1 ^a	1	6	—	8.2	136	4.2	64	4	48
2	1	6	—	4	28	3	83	1.5	0
3 ^a	1	6	—	6	95	3.5	32	5	23
4	1	3	—	2	6	1	2	1	2
5	1	6	—	8	41	4	40	3.5	23
6	3	4	—	2.6	15	1	0	4	0
7	1	5	Stable angina with persistent collaterals and occlusive disease in segments 1 and 2 of RCA requiring CABG	5	24	3.8	68	3.3	15
8 ^a	5	12	—	5.5	31	4	8	2.5	0
9	3	4	—	6	35	5	1	5.7	8
10	1	5	—	7	176	5.4	136	5.6	115
11 ^a	1	7	Sternal wound infection requiring debridement	6.5	92	5	75	3.5	15
Median (IQR)	1 (1-2)	6 (4.5-6)	Median	6	35	4	40	3.5	15

Abbreviations: AAOCA, anomalous aortic origin of a coronary artery; CABG, coronary artery bypass grafting; DC, discharge; ICU, intensive care unit; ME, morphine equivalents; POD, postoperative day; RCA, right coronary artery.

^aProcedure converted to partial or median sternotomy.

postoperative sternal precautions. While these endpoints were met for 6 of the 11 patients included in our early pilot, 4 patients required conversion to a partial or median sternotomy and 1 patient ultimately underwent CABG through a median sternotomy on POD 335. Despite performing over 300 index congenital cardiac cases annually during this time and having the operative assistance from our adult cardiac surgeons, who are familiar with this approach, our outcomes were less than ideal. That said, we believe the high conversion rate in this series is largely due to patient selection and not a technical learning curve that would improve with time or additional cases. In our experience, the ideal candidates for minimally invasive AAOCA repair are those with BMI <30 who are candidates for coronary unroofing, neo-ostial creation, or pulmonary artery translocation. Although, the anatomy dictates the final repair, and it is not fully appreciated until it is visualized following transverse aortotomy.⁷ This is different from the almost standardized approach associated with aortic valve replacement in adult cardiac surgery. Therefore, we would caution surgeons on generalizability of this approach for patients with BMI >30 or those for whom coronary reimplantation will likely be performed.

A limitation of this study is the small sample size and limited postoperative imaging. The majority of patients undergoing AAOCA repair in this study were referrals from outside hospitals. Following the first postoperative check, a majority of them were followed outside of our system. For this reason, we only have postoperative imaging on half of the patients, thus lacking objective data to support the subjective resolution of symptoms for all those included.

As congenital heart surgery becomes safer and patients live longer, the push for minimally invasive techniques will continue to grow. Our series shows that AAOCA repair through an anterior minithoracotomy is possible. Although given the high conversion rate, minimally invasive AAOCA repairs should be performed through a partial sternotomy, until the limitations of preoperative imaging improve.⁷ Early trends suggest that this approach may improve cosmetic outcomes and avoid sternal precautions for patients with BMI <30 who do not require coronary reimplantation. If these criteria are not met, surgeons performing AAOCA repair through an anterior minithoracotomy should be prepared for conversion to partial or median sternotomy.

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Declaration of Conflicting Interests

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